

(1) [10 points]

When you learned about A^* search, it was emphasized that the distance heuristic h must be an *underestimating* heuristic. Show us why, by giving an example of a graph with non-negative edge lengths, and some heuristic h that you get to design (that is not underestimating), such that A^* search does not find the shortest s - t path when using your heuristic h . Explain what the shortest path should be, and what A^* search finds and why.